AICTE VERIFIED STUDENT REGISTER

Role Features and Button Inventory

Detailed screen/control specification: purpose, preconditions, action, resulting state and permission boundary for every proposed button.

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Audience: AI coding agent + project owner

Document-control note
This is an implementation specification for a synthetic-data prototype, not an official AICTE standard, production approval, legal opinion, or proof of live government integration.

Source discipline: requirements from the PS 26263 implementation report and Sigma Batch curriculum are distinguished from proposed engineering choices. Any item that depends on a real policy, approval, API, credential trust chain, or deployment environment must be confirmed rather than guessed.

1. How to read this document

The original report defines the eight user types and broad workflows, but it does not prescribe exact button captions. The labels and button inventory below are PROPOSED UI requirements to make the MERN prototype implementable. Preserve these behaviours unless the project owner approves a change. A button must never imply a feature that is not implemented.

Button rule
Every state-changing button must have a loading state, success feedback, failure feedback, disabled/precondition state and backend authorization. Hiding a button does not replace server-side permission checks.

2. Shared public and authenticated controls

Screen / button	Action and conditions	Result / error behaviour
Login: Sign in	Validate fields; submit credentials once.	On success navigate by role. On failure show generic error; never expose server stack trace.
Login: Show password	Toggle only current password input visibility.	No server request.
Global: Sign out	POST logout endpoint; clear client user state after server response.	Return to login; if server fails, display status rather than claiming confirmed logout.
Global: Refresh	Reload current page data from authorised APIs.	Keep filters if practical; show retryable error.
Global: Search / Filter / Clear filters	Filter already-authorised rows by allowed fields.	Must not widen the caller's server-side scope.
Public verify: Scan QR	Request camera permission and decode a QR payload.	If permission denied, allow image/manual reference path only if implemented.
Public verify: Upload QR image	Accept an image under a configured size/type limit and decode.	Do not store image longer than necessary; reject malformed files.
Public verify: Verify	Call minimal public status endpoint for the credential reference.	Show VALID / REVOKED / INVALID-OR-TAMPERED only when supported by actual checks; avoid personal-data disclosure.
Public verify: Clear	Clear QR payload/result from UI.	No data mutation.
Global: Export	Download current permitted dataset/report.	Enforce scope and field masking on the server; log sensitive exports where appropriate.

3. Role 1 - College Operator (Maker)

Default scope: exactly one assigned institution. The operator can prepare and correct staged data, but cannot certify or commit official records.

Screen / button	Precondition and action	Expected result / blocked case
Dashboard: Upload batch	User has COLLEGE_OPERATOR role and assigned institution.	Open upload wizard; no access to another institution by URL manipulation.
Upload wizard: Download template	Download current demo template version.	CSV template and version note download; do not imply official live template sync.
Upload wizard: Choose file	Select CSV allowed by configured type/size.	Show file name and size; reject disallowed file and malformed extension/content.
Upload wizard: Remove file	Before submission only.	Clear selected file from UI.
Upload wizard: Pre-check	Run basic client checks; server still validates authoritatively.	Show missing columns/format issues without treating client checks as final verification.
Upload wizard: Submit upload	Valid file selection and allowed upload size.	Create batch and show batch ID/status; duplicate submission must be handled idempotently or explicitly rejected.

Screen / button	Precondition and action	Expected result / blocked case
Batch list: View batch	Batch belongs to user's institution.	Show status, counts, timestamps and permitted metadata.
Batch detail: Refresh status	Batch is visible to caller.	Fetch latest batch counts/progress.
Issues grid: Filter severity	Rows returned are already institution-scoped.	Filter Error/Warning/Pending/Info without hiding unresolved blockers from batch summary.
Issues grid: Search rows	Search permitted row fields.	Do not send raw sensitive ID to analytics/logs.
Issues grid: Edit row	Batch is uncommitted and user has edit permission.	Open editable allowed fields; immutable/internal fields stay read-only.
Issues grid: Save correction	Validate field permissions and persist correction to staging only.	Record correction metadata; do not change official registry.
Issues grid: Cancel edit	Discard unsaved local changes.	Return to read mode; confirm only if unsaved edits would be lost.
Issues grid: Revalidate changed rows	Saved corrections exist.	Re-run relevant checks; preserve previous issue history if required by audit design.
Issues grid: Export issues	Batch belongs to institution.	Download issues CSV with row numbers/rule codes; mask sensitive fields.
Seat ledger: View	Institution scope matches caller.	Read-only category capacity/filled/remaining counts from configured rules.
Conflicts: Open case	Case is assigned/visible to institution.	Show only permitted case fields; allow institution-side response/action only.
Batch detail: Mark ready for review	No blocking errors remain; required warning dispositions are recorded.	Set status to READY_FOR_REVIEW; this is not certification and must not commit records.
Certification control	Not available to operator.	No button and API returns 403 if directly invoked.

4. Role 2 - College Head / Nodal Officer (Checker)

Default scope: assigned institution only. The head is the maker-checker approver for an institution batch.

Screen / button	Precondition and action	Expected result / blocked case
Dashboard: Pending submissions	Load batches for assigned institution.	Show awaiting review, returned and certified counts.
Batch review: Open summary	Batch belongs to assigned institution.	Show row counts, error/warning dispositions, seat ledger impact and declaration.
Batch review: View issue details	Inspect unresolved and resolved issues.	Show evidence and clear rule descriptions; no editing of raw official fields by default.
Batch review: Return for correction	Batch not committed. Require a reason.	Set RETURNED_FOR_CORRECTION and notify operator in-app; no commit.
Batch review: Accept warning disposition	Only for warning types whose policy explicitly allows a documented disposition.	Store reason/actor/time; never suppress blocking errors or invent policy.
Batch review: Declaration checkbox	User confirms the displayed declaration before certification.	Required confirmation before enabling certify.
Batch review: Request OTP / verify OTP	Only if OTP is implemented for the prototype.	If not implemented, do not render a fake functioning OTP control; use clearly labelled mock confirmation only.
Batch review: Certify and submit	Correct institution, authority valid, no blocking errors, policy-required warnings resolved.	Server rechecks state and permissions, then commits safely; return conflict/validation error if conditions changed.
Batch review: Cancel	Before submission.	Return without mutation.
Conflicts: View case	Case within own institution.	Show other-institution identity only to minimum scope allowed.
Conflicts: Submit resolution	Choose permitted state/action and enter justification.	Record case event; update enrolment/seat ledger atomically if required.

Screen / button	Precondition and action	Expected result / blocked case
Reports: Export submission summary	Assigned institution only.	Download summary with masked sensitive fields.
Audit: View own submissions	Permission is limited to institution's actions.	Read-only timeline of changes/certification.

5. Role 3 - AICTE National Officer

Screen / button	Precondition and action	Expected result / blocked case
National dashboard: Refresh	National officer role.	Reload national aggregate metrics.
Filters: Academic year / state / status	Use explicitly approved filter values.	Update charts and tables; raw identifiers remain masked by default.
Map/KPI: Drill down	Click a state, institution or course.	Navigate one level deeper within national scope; preserve scope breadcrumbs.
Risk queue: Open case	Case visible in national scope.	Show score, reasons, evidence references, model/ruleset version and history.
Risk queue: Assign reviewer	Select authorised officer and record assignment.	Create assignment event and notification; reject invalid/out-of-scope assignee.
Risk case: Add note	Officer authenticated and case visible.	Append note with actor/time; do not overwrite the evidence.
Risk case: Mark reviewed	Only after required note/review fields are supplied.	Set review state; this does not imply fraud finding.
Risk case: Mark false positive	Requires reason.	Record feedback for future tuning; do not delete original signal.
Risk case: Escalate	Case open and reason supplied.	Create escalation event and notify next responsible authority.
Conflict monitor: Open/escalate	Case within national permissions.	Show conflict history and SLA status; no silent direct modification of college data.
Reports: Export national report	Only fields allowed for national role.	Export aggregated/masked information; record export event if sensitive.
Audit explorer: Verify chain	Audit service available.	Show pass/fail, affected sequence and verification time; never claim pass if not actually checked.
Policy controls	Not a routine national-officer button unless separately granted.	National monitoring role does not automatically grant system-admin rule-publishing powers.

6. Role 4 - AICTE Regional Officer

Screen / button	Precondition and action	Expected result / blocked case
Regional dashboard: Refresh	Officer has one or more assigned region IDs.	Return metrics for assigned regions only.
Filters: institution/course/year	All filters are constrained by server-side region scope.	Cannot reveal other regions by editing query parameters.
Institution row: Open details	Institution belongs to assigned region.	Show allowed aggregate and case details; identifiers masked by default.
Risk queue: Open / assign / add note	Case belongs to assigned region.	Support regional review workflow; assignee must be within permitted scope.
Risk queue: Mark reviewed / false positive	Reason required where relevant.	Record review; do not create an automatic penalty or fraud determination.
Conflict monitor: Open / escalate	Conflict falls within assigned region.	Read case and escalate to national level when required.
Regional report: Export	Filters constrained to assigned regions.	Produce report with no unrelated regional data.
National map: View all regions	Not permitted.	The UI should not show unrestricted national detail; direct API attempts return 403.

7. Role 5 - AICTE System Administrator

Screen / button	Precondition and action	Expected result / blocked case
Users: Create user	Required fields valid; role allow-listed; institution/region scope supplied when required.	Create inactive or invitation-pending account according to implemented onboarding; audit creator.
Users: Save changes	Validate unique email, valid role and scope.	Persist allowed changes; prevent user from assigning themselves higher privilege through a crafted request.
Users: Disable / re-enable	Admin confirms action and provides reason.	Disabled users cannot authenticate or continue protected sessions.
Users: Initiate password reset	User exists and reset mechanism is implemented.	Generate controlled reset workflow; never reveal/set/retrieve existing password.
Policies: Create draft	Rule set has academic year/version and data source metadata.	Save draft only; does not affect active validation.
Policies: Validate rules	Run schema/consistency tests on draft.	Show invalid config and test results; block publication on errors.
Policies: Compare versions	Select two stored versions.	Show differences and metadata/source notes.
Policies: Submit for approval	Draft valid and authored by current user.	Record request for a different authorised approver.
Policies: Approve/publish	Approver differs from rule author and has permission.	Publish a new version and retain previous version; audit actor/source/effective date.
Audit explorer: Verify chain	Audit store available.	Run integrity verification and display actual result.
System settings: Save	Only explicitly documented demo settings.	Do not expose secrets in UI or logs; every security-relevant change is audited.
Direct batch certification	Not permitted by this admin role alone.	Return 403 unless a distinct, explicitly approved additional role is assigned.

8. Role 6 - Degree-Awarding Body

Screen / button	Precondition and action	Expected result / blocked case
Certificate workspace: Search eligible student	Query restricted to issuer's assigned institution/records and permitted fields.	Return masked matching records only; no open-ended national student search.
Student row: View permitted details	Record is in issuer scope.	Show the minimum fields required to issue the credential.
Credential: Generate	Student record is eligible and issuer authorised.	Create a prototype credential reference/status; record issuer and timestamp.
Credential: Download QR/certificate	A generated credential exists.	Download a clearly labelled demo credential; do not claim cryptographic authenticity unless signed validation is implemented.
Credential: View status	Credential belongs to issuer scope.	Show active/revoked/invalid state from persisted data.
Credential: Revoke	Issuer has explicit authority; reason required.	Revoke only its own permitted credential and record audit event.
Certificate list: Export	Only fields required for the issuer's purpose.	Apply masking and authorization before export.

9. Role 7 - Verifier

Screen / button	Precondition and action	Expected result / blocked case
Verifier onboarding: Submit organisation	MVP may use a seeded, pre-approved organisation; do not claim real KYC integration.	Pending/approved state recorded according to prototype workflow.
New request: Select student	Enter permitted synthetic student reference.	Do not expose record details merely because a reference is known.
New request: Select purpose	Choose one of documented demonstration purposes.	Purpose is required and persisted.
New request: Select fields	Choose only allowed fields for that request type.	Server enforces allow-list; sensitive fields cannot be added by tampering with request body.
New request: Set expiry	Choose a permitted duration from configured options.	Expired request cannot be used later.
New request: Submit	Verifier account active and input valid.	Create pending request and notify student through demo notification UI.
Request detail: Refresh status	Request belongs to caller.	Show pending/approved/denied/expired/revoked status.
Result: View approved result	Consent is valid or explicitly configured legal basis exists. Default MVP: consent required.	Return only approved fields and result metadata.
Request: Cancel	Only own pending request.	Cancel pending request; no access to student data.
Bulk verify	Not included by default.	Do not render or enable unless legal basis, role scope and safeguards are approved.

10. Role 8 - Student

Screen / button	Precondition and action	Expected result / blocked case
My record: View	Authenticated student is mapped to a studentRef.	Show own enrolment and status only; no student search endpoint.
My record: Request correction	Select an allowed correction type and explain evidence.	Create correction request; official record remains unchanged until reviewed.
Correction: Submit request	Required explanation and any permitted evidence present.	Record request status and notify responsible institution user.
Consent inbox: Open request	Request refers to this student and has not expired.	Show requester, purpose, fields, expiry and prior status.
Consent: Approve	Request still pending; fields/purpose are displayed.	Record consent with approved fields and expiry; generate a signed consent artefact only if actual signing exists; otherwise label mock.
Consent: Deny	Request pending.	Record denial; verifier receives no requested private data.
Consent: Revoke	Consent is active and revocation is permitted.	Mark revoked; future access checks fail and access history records the event.
Access log: View details	Student's own access log only.	Show verifier, timestamp, purpose and fields disclosed; avoid exposing internal secrets.
Credential: Generate/download	Credential exists or issuance is enabled for student.	Provide QR/download with status and limitations stated.
Credential: Refresh status	Credential reference belongs to student.	Display current status from verification service.
Official record: Edit directly	Not permitted.	Use the correction-request workflow; direct mutation must return 403.

11. Shared button-state rules

- Disable double-submit while a mutation is pending.
- Require confirmation for destructive/high-impact actions such as user disablement, credential revocation, certification and rule publication.
- Display the exact blocking reason when a button is disabled.
- After mutation, refresh server state rather than trusting client-side guesses.
- Do not show controls the role cannot use; still enforce authorization on the server.
- Use plain-language errors with a field/row reference, for example: “Row 214: identifier format is invalid.”
- Do not create a button for a feature that is not wired to a real API or clearly labelled demo action.

12. Button/API traceability

Before coding a button, the agent must map it to: (a) role and scope, (b) preconditions, (c) API method/path, (d) server service, (e) database mutation/read, (f) audit event if relevant, (g) success UI state, and (h) failure tests. Buttons without a complete mapping are not ready to implement.